Frontier



Livermore Lab Foundation

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Livermore Lab Foundation

Organization location (we welcome applicants from anywhere in the world)

Livermore, CA, USA

Name(s) of primary point(s) of contact for this application

Livermore Lab Foundation (LLF): Sally Allen
Lawrence Livermore National Laboratory (LLNL): Briana Schmidt, Kari Finstad, Simon Pang

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

Non-profit organization

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Unlocking Stranded Carbon Assets

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Surface carbon mineralization typically uses crushed rock from mine tailings, industrial waste, or purpose-mined sources. The fastest reactions at ambient CO_2 (~400 ppm) occur in minerals such as brucite ($Mg(OH)_2$) and wollastonite ($CaSiO_3$), but these minerals are uncommon and often present at low concentrations. More abundant minerals, including olivine and serpentine, mineralize slower under ambient conditions, in part because CO_2 supply is limited. Higher temperature and pressure, or pure CO_2, can greatly increase reaction rates, but the added energy use can partially or fully offset net carbon removal (Power et al. 2013). Lab studies and modeling suggest a nonlinear relationship between pCO_2 and mineralization rate, indicating that even a modest enrichment (5% to 10% CO_2) may approach rates seen under pure CO_2 for some systems (Harrison et al. 2013; Hamilton et al. 2020; Pasquier et al. 2014). These CO_2 concentrations are consistent with flue and exhaust gas from other carbon dioxide removal (CDR) pathways such as mobile pyrolysis units or could be achieved with low-purity direct air capture (DAC). However, data for common minerals are limited and the controlling mechanisms remain unclear. Moreover, most prior work is benchtop-scale, and field trials are needed to advance the science and enable climate-relevant scale-up (Kelemen et al. 2019). Our project will test the hypothesis that modest CO_2 enrichment will improve mineralization kinetics sufficiently to compete economically with other CDR options. We will evaluate common rocks and minerals, measure kinetics in the lab, validate results in a small-scale field trial, and complete a high-resolution techno-economic analysis (TEA) for California.

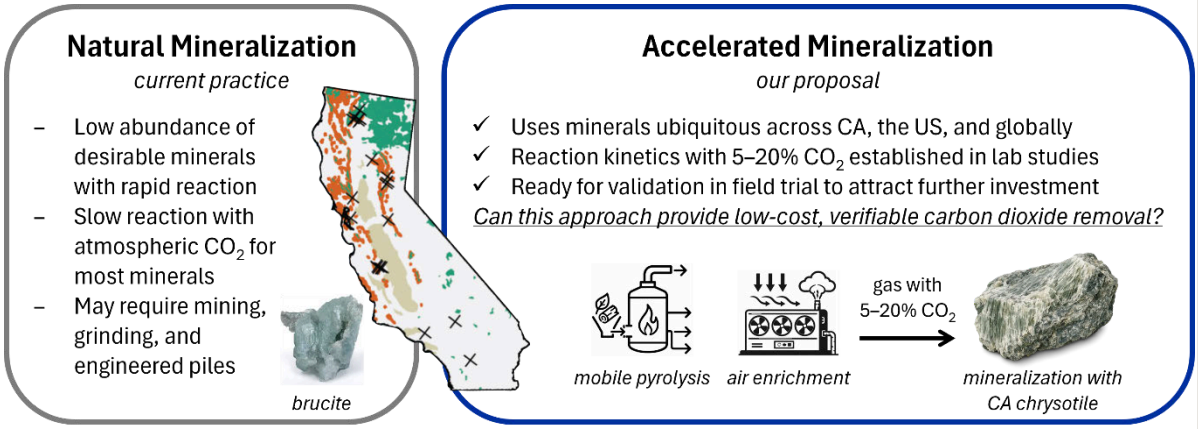


Figure 1. California has abundant rock resources (orange and green areas) for accelerating carbon mineralization, potentially using dilute CO_2-bearing gases. Our project will leverage LLNL's new wet rock chemistry lab facilities to measure crucial mineralization rate data, conduct a field validation trial, and analyze the geospatial and technoeconomic potential for using these minerals for low-cost and verifiable CDR.

Broader impacts on the field include:

i. Measuring kinetics for the most common rocks and minerals under consistent and comparable conditions and identifying rate-limiting steps.

ii. In the use-case of pairing with mobile pyrolysis units: creating a local solution for dispersed

biomass feedstock (e.g. products of forest management) and improving CDR efficiency by capturing exhaust CO_2 with mobile mineralization reactors. This can also support rural economic development and energy security in communities with abundant biomass feedstock and mineralization-suitable rocks, for example in Northern CA where LLNL has existing collaborations with the Karuk Tribe and Sierra Institute for Community and Environment.

iii. In the use-case of pairing with low-purity DAC: reduce energy, water demand, and cost relative to traditional DAC and broaden siting options and accelerate mineralization versus ambient air, while retaining all the monitoring, reporting, and verification (MRV) benefits of traditional DAC and mineralization.

iv. Providing field validation of laboratory results, addressing a widely cited need for field trials to advance science and enable scale-up of climate technologies.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

Our project will test whether geographically stranded carbon can be made economically viable by pairing low-purity CO_2 with carbon mineralization. California is an ideal place to test these concepts because it has:
a. Vast surface exposures of ultramafic and mafic rocks suitable for carbon mineralization, including large quantities of mine tailings and waste rock (Figure 2A).
b. Abundant, diverse, and dispersed biomass feedstock, particularly in the northern part of the state where traditional saline geologic storage is unavailable (Figure 2B).

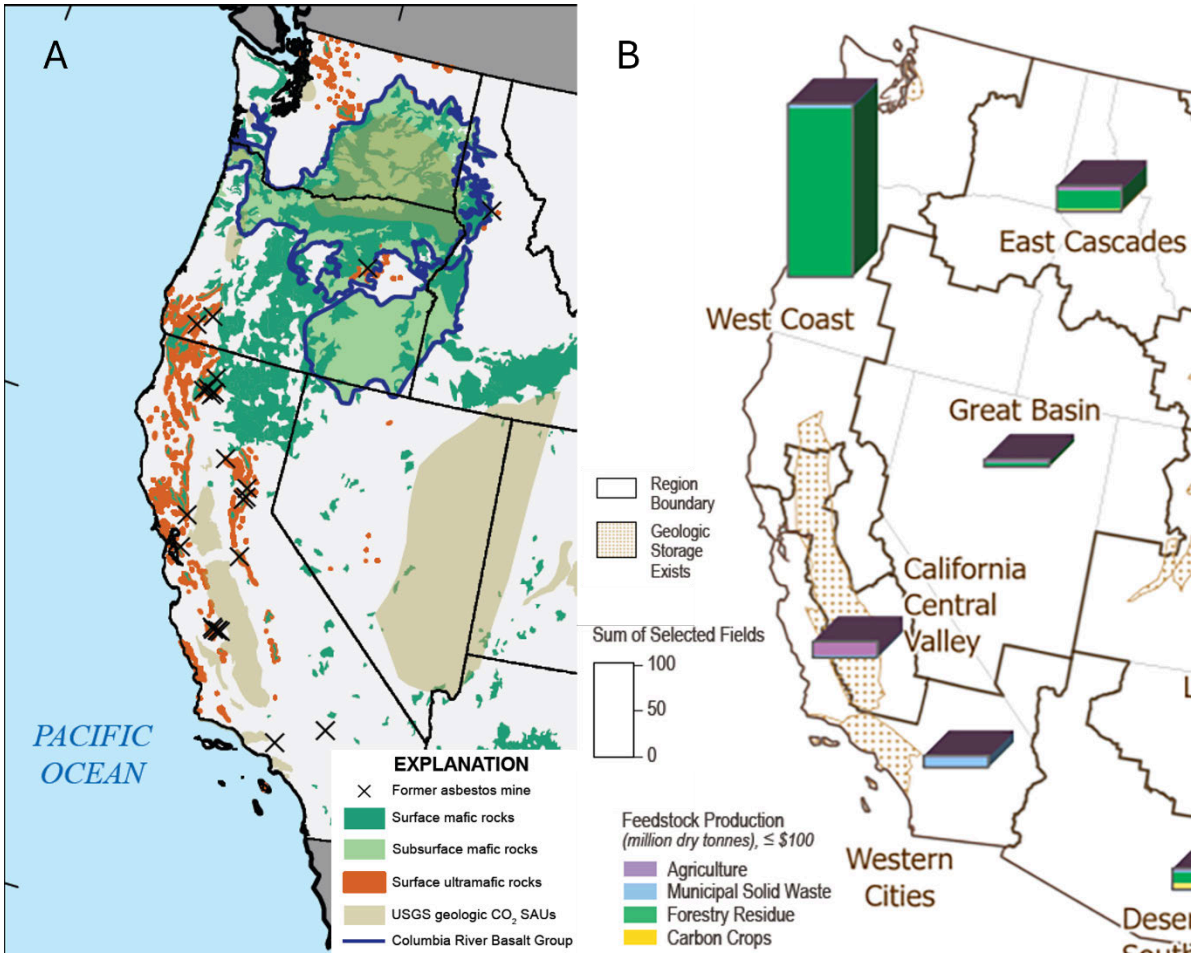


Figure 2. Example of stranded carbon assets: the West Coast has ample feedstock for biomass carbon removal (2B) but limited options for traditional geologic CO_2 storage (2A, tan). Mineral storage is abundant (2A, green and orange) but technologically immature. (Sources: A. Blondes et al., 2019; B. Pett-Ridge et al., 2023)

We will address three key challenges:

i. Unlock abundant rocks and minerals. We will screen California geology using existing maps and geochemical datasets to identify high-potential regions, common rock types, and mineral assemblages. We will then collect representative samples and run lab experiments in LLNL's new wet-chemistry facility using 1) CO_2 concentrations consistent with low-purity streams (approximately 5 to 20%), and 2) ambient CO_2 concentrations, for rate comparison. We will vary parameters to simulate different CO_2 sources to aid in identification of rate limiting factors, for examples elevated temperature and water content consistent with mobile pyrolysis exhaust, and measure carbon mineralization reaction rates at both elevated and ambient CO_2 concentrations.

ii. Field validation. We will conduct a field test at the closed King City Asbestos Corporation (KCAC) mine site to validate laboratory results. Using off-the-shelf materials and apparatus, we will assess mineralization rates under conditions as close as possible to the lab-identified optimum, using approximately 1 metric ton of mine waste rock. We will finalize the design early in Phase 2 based on laboratory results and site constraints. Holding all other variables constant, experiments will be

performed at both ambient atmospheric and elevated CO_2 concentrations to allow direct comparison of rates.

iii. Cost. Previous work has assessed the economics and removal potential of surface mineralization under ambient pCO_2, temperature, and pressure conditions or under elevated temperature and pressure conditions. However, the economic competitiveness of combining low-purity CO_2 with surficial mineralization under otherwise ambient conditions, relative to conventional removal, capture, and geologic storage, remains largely untested at scale. We will conduct a high-resolution, geospatial techno-economic analysis using California-specific mineral and CO_2 resources and real-world constraints to determine feasibility and key cost drivers. This will be informed by both the lab and field experiments, allowing us to directly compare and cost trade-offs associated with concentrated CO_2 streams versus ambient atmospheric CO_2.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

The project faces several identifiable risks, each with a defined mitigation strategy. A low risk is that field testing may not be achievable within the available time and budget. However, our existing site access and prior work at the location substantially reduce this risk, and we do not anticipate needing additional approvals beyond modifications to our EH&S protocols. A moderate risk is that low-purity CO_2 alone may not be sufficient to overcome mineralization rate limitations, especially for more abundant but slower-reacting rocks and minerals. To address this, we will examine the rate-limiting steps associated with the most common rock and mineral types and develop strategies to overcome them, including approaches such as biomineralization (Power et al. 2016; McCutcheon et al. 2016). These analyses will also help constrain uncertainty related to surface mineralization removal and storage capacity, thereby informing climate models and broader assessments. A high risk is that some approaches may be relevant only at the kiloton scale rather than the gigaton scale, which could limit their contribution to climate goals. We will mitigate this by evaluating how insights from these processes can be translated into commercial-scale applications and by identifying economically compelling use cases, such as commercial-scale biomass carbon removal paired with large-scale surface mineralization storage.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	Map and database of locations for sample collection; Rock samples and geochemical analyses incl. whole rock major and trace elements (XRF, ICP-MS), mineralogy (XRD, TGA-MS, SEM) and total carbon; Raw data and fully analyzed and interpreted results

	from lab experiments; Description of field test design; Preliminary TEA whitepaper; approved work control document for field experiments; Raw data, geochemical analyses, and fully analyzed and interpreted results for field experiment; Final TEA whitepaper; Completed manuscripts for full project
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	Linking low purity CO_2 streams with carbon mineralization using abundant rock types could unlock major new pathways to contribute to carbon removal goals. Focusing on increasing the reaction rate of high-abundance slow-reacting minerals without requiring increased pressure or temperature or extensive pretreatment will provide a compliment to current surface mineralization approaches, which mainly rely on low-abundance fast-reacting minerals.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	Phase 1, 8-12 Months: Identify high potential regions in CA, All rock samples collected and prepared for experiments, Experiments commence, Complete lab experiments and data analysis; Phase 2, 8-12 Months: Finalize design of field test and prepare for fieldwork, Complete preliminary TEA, Complete fieldwork & analyses, Complete final TEA, Submit at least one manuscript, Hold a public briefing for interested stakeholders and partners in CA
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$750k